

Temporal Non-Constructibility in General Relativity: Lapse, Foliations, and the Operational Limits of Time

Asaad N. Riaz

Riaz Communications Inc.

Correspondence: asaad@quantumhub.solutions

General Relativity describes spacetime through arbitrary foliations by spacelike hypersurfaces, each labeled by a coordinate time t and associated lapse function $N(x, t)$. These quantities enter the ADM evolution equations, yet the theory contains no distinguished slicing or preferred temporal rate. This raises a fundamental question: can any internal observer operationally recover the global time parameter used in a foliation?

We address this by formulating and proving the *Relativistic Temporal Non-Constructibility Theorem*. Working strictly within classical Lorentzian geometry and the ADM $3 + 1$ decomposition, we model an observer as a timelike worldline with access only to events in its causal past. We show that for any two foliations related by a smooth, strictly monotonic reparametrization $t \mapsto t' = f(t)$, the lapse transforms as $N \mapsto N' = N (dt/dt')$, while the four-geometry, causal structure, and proper time along all worldlines remain invariant. As a consequence, every scalar observable accessible to an internal observer yields the same empirical history in both descriptions. The coordinate time t and lapse function $N(x, t)$ are therefore *gauge variables with no operational significance*.

This result clarifies several structural features of general relativity. It provides the operational foundation for the absence of global simultaneity, explains why the ADM Hamiltonian is a sum of constraints with $H \approx 0$, and shows that the timeless form of the Wheeler-DeWitt equation reflects temporal gauge freedom rather than a failure of quantization. The theorem generalizes the classical Temporal Non-Constructibility Theorem for autonomous dynamical systems, establishing that global time parameters are unobservable in both Newtonian and relativistic settings.

Proper time along worldlines, the geometric invariant determined by the metric, is the only empirically meaningful temporal quantity in general relativity. Global coordinate time plays no physical role beyond labeling slices of a foliation. This is the content of temporal non-constructibility in GR.

I. INTRODUCTION

The description of physical evolution in general relativity relies on a foliation of spacetime by spacelike hypersurfaces. In the ADM formulation, [1] a spacetime (M, g_{ab}) is decomposed into slices Σ_t labeled by a real parameter t , and Einstein's equations are expressed as evolution equations for geometric data on these slices. The lapse function $N(x, t)$ determines the amount of proper time between adjacent hypersurfaces, while the shift vector $N^i(x, t)$ encodes how spatial coordinates are transported between them. Although this construction provides a convenient Hamiltonian representation of the theory, the parameter t itself is not part of the physical geometry. It merely labels members of the chosen foliation.

This raises a fundamental question:

Does the coordinate time t used in a foliation have operational significance for an internal observer, or is it purely a gauge artifact?

In the standard formulation of general relativity, no structure intrinsic to the metric singles out any particular foliation or any particular rate of advance of the slicing parameter. Different choices of lapse correspond to different ways of assigning labels to the same family of hypersurfaces. Because the geometry of spacetime is invariant under diffeomorphisms, and because observers are represented as worldlines within that geometry, it is not clear whether any observer can extract information

about which foliation or which lapse function was used to generate an ADM decomposition.

A related question arises in classical autonomous dynamical systems. There, the Temporal Non-Constructibility Theorem (TNT) [2] demonstrates that the external time parameter t appearing in the equations of motion is not an observable quantity: reparametrizations of the form $t \mapsto f(t)$ leave all observationally accessible histories invariant. The classical theory thereby distinguishes the geometric orbit traced by the system from the arbitrary parameter used to describe its traversal.

The present work establishes the precise analogue of this result in general relativity. *We prove that the foliation-time coordinate t and the lapse function $N(x, t)$ are operationally non-constructible*: no internal observer, modeled as a timelike worldline with access only to events in its causal past, can measure or infer the global slicing parameter used in an ADM formulation of the spacetime. Proper time along worldlines is invariant under any smooth reparametrization of the foliation-time coordinate, and the observer's recorded history of scalar observables is unaffected. Consequently, all physical predictions of the theory are independent of the choice of lapse.

This result explains, in operational terms, several structural features of general relativity: the absence of a preferred notion of global simultaneity, the vanishing of the canonical Hamiltonian in the ADM formalism, and the timeless character of the Wheeler-DeWitt equation [3]

in canonical quantum gravity. It shows that the non-observability of global time in GR is not an interpretive issue but a mathematical consequence of the diffeomorphism invariance of the theory and the locality of observation.

The structure of the paper is as follows. Section II develops the relativistic framework, introducing Lorentzian geometry, foliations, and the ADM decomposition. Section III formulates the observer model and defines observable histories. Section IV analyzes temporal reparametrizations of the foliation and their effect on lapse, proper time, causal structure, and observable histories. Section V states and proves the Relativistic Temporal Non-Constructibility Theorem. Section VI explores its implications for simultaneity, Hamiltonian evolution, and canonical quantum gravity. Section VII discusses limitations, and Section VIII summarizes the conclusions.

General relativity provides a rich geometric structure, but none of that structure defines a measurable global time. The only invariant temporal quantity is proper time along worldlines. The aim of this paper is to show this rigorously, using only the standard tools of Lorentzian geometry and ADM dynamics, and to clarify the operational meaning of time in relativistic physics.

II. RELATIVISTIC FRAMEWORK

This section establishes the geometric and dynamical structures required for the relativistic formulation of temporal non-constructibility. The aim is to work entirely within the standard setting of classical general relativity, without introducing additional assumptions or structures beyond those ordinarily used in the ADM treatment of gravitational dynamics. All relevant concepts, spacetime, observers, foliations, and lapse functions, are defined explicitly so that the subsequent analysis proceeds in a simple and systematic manner.

A. Spacetime as a Lorentzian Manifold

A spacetime is specified by a smooth, connected, four-dimensional manifold M equipped with a smooth Lorentzian metric g_{ab} of signature $(-+++)$. No preferred coordinates or global time functions are assumed. The geometry supplies a light-cone structure that distinguishes timelike, null, and spacelike directions at each point.

For the purposes of formulating the temporal non-constructibility theorem, we assume that spacetime is globally hyperbolic.[4, 5] This ensures that well-posed initial-value formulations exist: every solution of Einstein's equations is uniquely determined by initial data on a spacelike hypersurface, and every causal curve intersects each hypersurface exactly once. This condition is standard in the ADM formulation and guarantees the existence of foliations by Cauchy surfaces.

No foliation is regarded as physically distinguished; each arises from a choice of coordinates and carries no operational content for any internal observer.

B. ADM 3+1 Decomposition and Foliation

To express the dynamics of general relativity in a form resembling an evolution system, one often introduces a foliation of spacetime: [1, 6]

$$M \cong \mathbb{R} \times \Sigma, \quad (1)$$

where each Σ_t is a smooth spacelike hypersurface labeled by a real parameter t . This construction is not intrinsic to the geometry; it is a choice of slicing, introduced for convenience in forming a Hamiltonian description.

Relative to such a foliation, the spacetime metric takes the usual ADM form:

$$ds^2 = -N^2 dt^2 + h_{ij}(dx^i + N^i dt)(dx^j + N^j dt). \quad (2)$$

Here: h_{ij} is the induced spatial metric on the slice Σ_t , N is the lapse function, determining the amount of proper time between adjacent slices, N^i is the shift vector, determining how spatial coordinates are displaced as one moves between slices.

Given initial data (h_{ij}, K_{ij}) on a slice Σ_t , where K_{ij} is the extrinsic curvature, Einstein's equations decompose into evolution equations for (h_{ij}, K_{ij}) and two sets of constraints: the Hamiltonian constraint and momentum constraints. The lapse function N and the shift vector N^i appear in these equations as freely specifiable multiplier fields. [1, 6]

The foliation and the parameter t are arbitrary; the theory contains no preferred slicing, and no physical observable depends on the choice of N or N^i .

C. Gauge Freedom of the Lapse

The lapse function $N(x, t)$ governs the rate at which the foliation parameter t advances in relation to the physical proper time [1, 7] measured along curves orthogonal to the slices Σ_t . Its value can be specified freely at each spacetime point. Different choices of N correspond to different ways of labeling the same geometric hypersurfaces, and hence to different coordinate descriptions of the same spacetime.

If one performs a smooth, strictly monotonic reparametrization of the foliation parameter,

$$t \mapsto \tau = f(t), \quad f'(t) > 0, \quad (3)$$

then the lapse transforms as

$$N \mapsto \tilde{N} = N \frac{dt}{d\tau}. \quad (4)$$

This transformation alters the coordinate representation of the evolution but does not affect any geometric

or physical quantities. [6] It is the relativistic analogue of the time-reparametrization freedom in autonomous dynamical systems.

The lapse is therefore a gauge variable: it reflects a freedom of description rather than a measurable aspect of the spacetime. Any two choices of lapse related by the transformation above produce identical physical predictions for all internal observers.

III. INTERNAL OBSERVERS IN GENERAL RELATIVITY

A. Observers as Timelike Worldlines

A physical observer in general relativity is represented by a future-directed timelike curve

$$\gamma : I \subset \mathbb{R} \rightarrow M, \quad (5)$$

with tangent vector

$$u^a = \frac{dx^a}{d\lambda}, \quad (6)$$

where λ is any smooth, strictly monotonic parameter along the curve. The geometric object is the image $\gamma(I)$; the particular parametrization carries no physical content.

No reference is made to the global foliation or to ADM time t . The observer is part of the spacetime manifold itself, not external to it. [4]

B. Proper Time as the Physical Temporal Parameter

Along γ , the invariant temporal structure is the proper time, defined by

$$d\tau = \sqrt{-g_{ab}u^a u^b} d\lambda. \quad (7)$$

Proper time depends only on the spacetime metric g_{ab} and the tangent to the worldline. It is:

- invariant under coordinate transformations,
- invariant under foliation changes,
- invariant under any reparametrization of ADM coordinate time.

Accordingly, τ provides the only observer-independent temporal parameter available to an internal observer. [4]

C. Observable Algebra Along a Worldline

The observer can evaluate any smooth scalar field $\phi : M \rightarrow \mathbb{R}$ at the events along its worldline. The collection of observable quantities accessible to the observer is therefore

$$\mathcal{O}_\gamma = \{ \phi(\gamma(\lambda)) : \phi \in C^\infty(M) \}. \quad (8)$$

Tensor fields are accessible only through scalar contractions with u^a or the metric; what matters is the value at the event on the worldline. No structure outside the immediate event on γ is assumed.

D. Causal-Past Accessibility

General relativity constrains the information available to an observer by causal structure. The observer can only receive signals from the causal past of its worldline, defined by

$$J^-(\gamma) = \bigcup_{\lambda \in I} J^-(\gamma(\lambda)). \quad (9)$$

Any field value outside $J^-(\gamma)$ is operationally inaccessible. [5] This assumption enforces relativistic causality and replaces any reference to “spatial information” or “global synchrony,” which have no invariant meaning.

E. Observable Histories and Their Parametrization

A finite set of observables measured along γ defines a parametrized history

$$\lambda \mapsto F(\gamma(\lambda)) \in \mathbb{R}^k, \quad (10)$$

where F collects the values of the fields under consideration.

Two parametrized histories describe the same observational content if they differ only by a smooth, strictly monotonic reparametrization $\lambda \mapsto \tilde{\lambda} = f(\lambda)$. Thus, the observer’s empirical history is captured by the equivalence class

$$[\gamma] \in \text{Path}(\mathbb{R}^k) / \text{Diff}^+(\mathbb{R}). \quad (11)$$

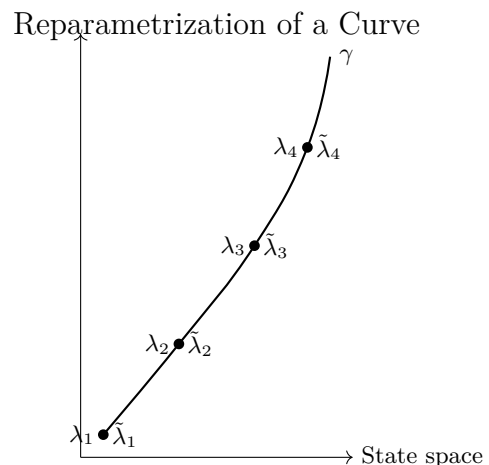


FIG. 1: A single geometric worldline admits multiple parametrizations. Two monotonic parameters λ and $\tilde{\lambda} = f(\lambda)$ trace the same curve and produce identical observable histories.

This construction isolates the physical content of observations from arbitrary choices of parameter. It mirrors the role of observable histories in the classical TNT theorem, now formulated within relativistic spacetime.

IV. TEMPORAL REPARAMETRIZATIONS IN GENERAL RELATIVITY

A. Reparametrizing the Foliation-Time Coordinate

In the ADM decomposition, the spacetime manifold M is foliated by hypersurfaces Σ_t labeled by a coordinate t .

A change in the labeling of these hypersurfaces is given by a smooth, strictly monotonic reparametrization

$$t \mapsto \tau = f(t), \quad f' > 0. \quad (12)$$

Under this transformation, the spatial metric h_{ij} and shift vector N^i transform as tensorial fields under coordinate change, while the lapse function rescales according to

$$N \mapsto \tilde{N} = N \frac{dt}{d\tau}. \quad (13)$$

This rescaling reflects the fact that the lapse controls the relative advance of proper time between adjacent hypersurfaces:

$$d\tau_{\text{proper}} = N dt = \tilde{N} d\tau', \quad (14)$$

Thus, a reparametrization of the foliation-time coordinate induces a corresponding gauge transformation of the lapse, leaving the geometric content of the spacetime unchanged. [1]

B. Proper Time Is Invariant Under Refoliation

For a timelike worldline γ , the proper time increment is given by

$$ds^2 = -g_{ab} dx^a dx^b, \quad d\tau = \sqrt{-g_{ab} u^a u^b} d\lambda. \quad (15)$$

Neither the coordinate t nor the lapse N appears in this expression.

Therefore, reparametrizing the foliation-time coordinate leaves the proper time experienced by the observer invariant:

$$d\tau \text{ is independent of } t, \tau, N, \tilde{N}. \quad (16)$$

Proper time is a geometric invariant of the metric and worldline alone; it carries all the physical temporal information available to the observer. [4]

C. Causal Structure Is Invariant Under Refoliation

A change in the foliating coordinate does not change the causal structure of spacetime. The chronological and causal relations determined by g_{ab} remain fixed:

$$J^-(\gamma) \text{ is identical before and after } t \mapsto \tau. \quad (17)$$

Accessible information, the fields restricted to the causal past, thus remains unchanged. [5] This invariance plays the exact role that phase-space orbit invariance played in the classical TNT setting.

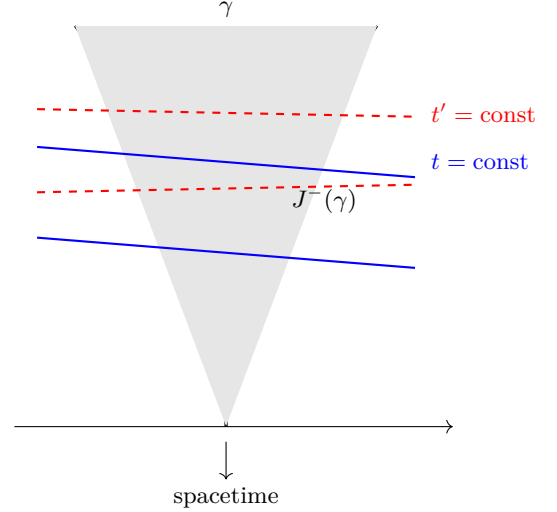


FIG. 2: Two distinct foliations $\{\Sigma_t\}$ and $\{\Sigma_{t'}\}$ of the same spacetime intersect the same causal past $J^-(\gamma)$ of an observer. Because proper time and causal structure depend only on the metric, no internal observer can determine which foliation labels the slices.

D. Observable History Is Invariant Under Temporal Reparametrization

Let $F(\gamma(\lambda))$ denote any observable history measured by the observer along its worldline.

Since the values of fields at spacetime points remain unchanged, and since the worldline itself is a geometric curve, a reparametrization of global time leaves the empirical record invariant:

$$F(\gamma(\lambda)) = F(\gamma(f^{-1}(\lambda))). \quad (18)$$

up to a reparametrization of the curve's parameter. Thus, the observable history satisfies

$$[\gamma] = [\gamma \circ f^{-1}] \quad \text{in} \quad \text{Path}(\mathbb{R}^k)/\text{Diff}^+(\mathbb{R}). \quad (19)$$

No internal observation distinguishes between the original foliation-time coordinate t and the reparametrized coordinate τ .

E. Summary

- Reparametrizing the foliation-time coordinate t induces a rescaling of the lapse function N .
- Proper time along any worldline is invariant under such changes.
- Causal structure and the observer's accessible domain remain unchanged.
- Observable histories are unaffected up to reparametrization.

These facts prepare the ground for the *Relativistic Temporal Non-Constructibility Theorem*, where we formalize the claim that *no internal observer can recover the coordinate-time slicing or the lapse function from physical measurements*.

V. THE TEMPORAL NON-CONSTRUCTIBILITY THEOREM (RELATIVISTIC TNT)

This section states and proves the central result of the paper. The theorem establishes that in general relativity, the global temporal parameter used to label slices of a foliation carries no operational significance for any internal observer. The lapse function and slicing parameter may be altered without altering the empirical content of the observer's experience.

A. Statement of the Theorem

Theorem (Relativistic Temporal Non-Constructibility).

Let (M, g_{ab}) be a smooth, globally hyperbolic spacetime. [5] Let $\{\Sigma_t\}$ and $\{\Sigma_{t'}\}$ be two smooth foliations of M related by a strictly monotonic reparametrization

$$t \mapsto t' = f(t), \quad (20)$$

with corresponding lapse functions related by

$$N \mapsto N' = N \frac{dt}{dt'}. \quad (21)$$

[1]

Let γ be any future-directed timelike worldline representing an internal observer. Then:

- The causal past $J^-(\gamma)$ is identical for both foliations.
- All physical observables recorded along γ take the same values at the same spacetime events.
- The observer's measured history lies in the same equivalence class of parametrized paths:

$$[\gamma]_{\{t\}} = [\gamma]_{\{t'\}}. \quad (22)$$

Conclusion. No internal observer can detect, reconstruct, or operationally distinguish the global temporal parameter t or the lapse function $N(x, t)$ used to generate a foliation of spacetime. Global coordinate time in GR is a gauge variable.

B. Proof of the Theorem

The proof has three components: slicing invariance, causal-past invariance, and observable-history invariance.

(i) Slicing Invariance

A reparametrization $t \mapsto t' = f(t)$ induces the lapse transformation

$$N \mapsto N' = N \frac{dt}{dt'}. \quad (23)$$

This rescaling changes the labeling of hypersurfaces Σ_t to $\Sigma_{t'}$ without altering the underlying four-geometry. The spacetime metric g_{ab} retains the same geometric structure, and all invariant geometric quantities, curvature, causal relations, and proper time, remain unchanged.

Thus the two foliations describe the same spacetime; only the coordinate labels differ.

(ii) Causal-Past Invariance

The causal past $J^-(\gamma)$ is defined purely in terms of the spacetime metric. Because the metric is unchanged under the reparametrization of the foliation parameter, the set of spacetime points that can influence γ remains identical. Hence the observer has access to the same information, fields, and events in both descriptions.

(iii) Observable-History Invariance

Let the observer's measured quantities be given by scalar fields evaluated along the worldline. Parametrize the worldline by proper time τ , a geometric invariant defined by

$$d\tau^2 = -g_{ab} dx^a dx^b. \quad (24)$$

[4]

The measurement record is

$$\Phi(\tau) = \phi(\gamma(\tau)). \quad (25)$$

Since τ depends only on the metric and the path, it is independent of the foliation parameter t , the transformed parameter t' , or the lapse functions N and N' . Therefore the sequence of values $\Phi(\tau)$ is identical under both foliations.

The observer’s empirical history therefore defines the same equivalence class of parametrized curves in observable space:

$$[\gamma]_{\{t\}} = [\gamma]_{\{t'\}}. \quad (26)$$

C. Conclusion of the Proof

Combining (i), (ii), and (iii), the observer’s accessible information, the causal structure of the spacetime, and the recorded sequence of observables remain unchanged under any smooth, strictly monotonic reparametrization of the foliation parameter. Thus the choice of global time coordinate, and its associated lapse function, is operationally non-constructible.

This completes the proof. $\square$

VI. CONSEQUENCES FOR GENERAL RELATIVITY

This section summarizes the implications of the Relativistic Temporal Non-Constructibility Theorem for the structure of general relativity. The results clarify several long-standing conceptual issues—particularly the absence of a physical global time, the role of the Hamiltonian constraint, and the timeless character of canonical quantum gravity.

A. Absence of an Observable Global Time

General relativity contains no preferred global temporal parameter. Coordinate time t enters only through a chosen foliation of the spacetime manifold and its associated lapse function $N(x, t)$. [1] Section V established that this parameter is operationally inaccessible to any internal observer. Proper time along worldlines is invariant; the foliation label is not.

Two immediate consequences follow:

- i. *Global simultaneity is not physical.*
No observer can determine whether two spatially separated events share the same value of t . [8]
- ii. *The “rate” at which the universe evolves cannot be measured.*
Rescaling t and N produces empirically identical observational histories.

Thus the only temporal quantity with invariant content in GR is the proper time τ along a worldline. All global time parameters are artifacts of coordinate choice.

B. Hamiltonian Constraint and the Vanishing of Energy

In the ADM 3+1 formulation, the Hamiltonian of general relativity takes the form

$$H = \int (NH + N_i H_i) d^3x, \quad (27)$$

where H and H_i are the Hamiltonian and momentum constraints. The lapse N and shift N_i appear as Lagrange multipliers enforcing these constraints. [1, 7]

The Relativistic TNT now provides the operational reason for this structure:

- N is a gauge variable.
- Changes in N correspond to reparametrizations of the foliation parameter t .
- Because N carries no observable information, the physical Hamiltonian must not generate genuine evolution.

Hence the condition

$$H \approx 0 \quad (28)$$

is not a mystery or pathology; it is the natural expression of temporal gauge freedom. The Hamiltonian of GR must vanish (up to constraints) precisely because the global time parameter is not a physical observable.

This resolves the conceptual tension often associated with “frozen-time” formulations of GR. The theory is not frozen; rather, physical evolution is encoded in geometric relations and proper time, not in coordinate time.

C. Wheeler-DeWitt Equation and Timelessness in Quantum Gravity

Canonical quantization of GR promotes the Hamiltonian constraint to the operator equation

$$H\Psi = 0, \quad (29)$$

the Wheeler-DeWitt equation. [3] The absence of a time derivative in this equation is traditionally interpreted as a profound conceptual puzzle: quantum gravity appears to be timeless.

The Relativistic TNT clarifies this directly:

- The classical theory contains no observable global time.
- Therefore, its quantum counterpart cannot contain a generator of evolution with respect to t .
- The equation $H\Psi = 0$ is the quantum expression of temporal gauge invariance.

Timelessness is not a failure of quantization; it is the inevitable quantization of a theory whose global temporal parameter is unobservable and gauge-dependent.

D. Relativity of Simultaneity Reinterpreted

Special Relativity already denies the physical meaning of global simultaneity. [8] The relativistic version of TNT generalizes this:

- No foliation of spacetime carries invariant operational significance.
- No experiment confined to a worldline can reveal which slicing was used in a global description.
- Einstein’s observation, “simultaneity is not a matter of experience”, is elevated from a kinematic fact to a structural invariant of all relativistic physics.

The inability to reconstruct a preferred slicing is therefore not a limitation of measurement, but a mathematical necessity.

E. Connection to Classical Temporal Non-Constructibility

The classical TNT established that for autonomous dynamical systems on smooth manifolds, the external parameter t appearing in the equations of motion is a gauge variable: unobservable and operationally non-constructible. [2]

The relativistic theorem extends this result:

- **Classical TNT:** reparametrizing t leaves all observable histories invariant.
- **Relativistic TNT:** reparametrizing the foliation time t and lapse N leaves all observable histories invariant.

Together these results show:

- *Time, as a global parameter, is a gauge degree of freedom in all classical physics.*
- *Only proper time and observable relational structure carry empirical significance.*

VII. LIMITATIONS

The results established in this paper concern the operational status of global time in classical general relativity. While the Temporal Non-Constructibility Theorem provides a rigorous characterization of temporal gauge freedom within the ADM framework, its scope is limited in several important respects. These limitations clarify the domain of validity of the theorem and delineate the boundary for future extensions.

A. Restriction to Classical General Relativity

The analysis is confined to smooth Lorentzian spacetimes satisfying Einstein’s field equations. Quantum effects, including those associated with quantum matter fields or attempts to quantize the spacetime metric itself, are not considered. The theorem does not address quantum fluctuations of geometry or the behavior of temporal gauge at the Planck scale. These issues are reserved for subsequent work extending the framework to semiclassical and canonical quantum gravity.

B. Requirement of Global Hyperbolicity

The ADM decomposition assumes that the spacetime is globally hyperbolic. This ensures the existence of a foliation by Cauchy surfaces and allows one to formulate initial-value evolution equations in a well-defined way. [4, 5] The theorem therefore does not apply to:

- spacetimes with closed timelike curves,
- non-globally hyperbolic geometries such as Anti-de Sitter space without appropriate boundary conditions,
- or pathological solutions lacking a Cauchy surface.

The conclusion that global time is unobservable holds only within the class of spacetimes where foliation-based dynamics is meaningful.

C. No Claim About the Existence or Non-Existence of Time

The theorem does not assert that “time does not exist,” nor does it imply that physical evolution is illusory. What is shown is narrower and fully operational:

- the foliation parameter t used to slice spacetime is not accessible to any internal observer,
- the lapse function $N(x, t)$ is a gauge variable,
- and physical time corresponds to proper time along worldlines.

The theorem concerns the non-constructibility of the coordinate label, not the physical processes that unfold along timelike paths.

D. No Implications for Discrete or Non-Manifold Models

The argument relies on smooth manifold structure, differentiability of fields, and the causal properties of Lorentzian geometry. The theorem does not directly apply to:

- causal set models,
- spin-foam or loop quantum gravity formulations,
- discrete lattice models,
- or non-Riemannian geometries.

Extensions to such frameworks would require reinterpreting “slicing,” “lapse,” and “observer” in non-smooth settings.

E. Dependence on the Standard Observer Model

The observer is modeled as a single timelike worldline with access to fields in its causal past. This is the standard local observer concept in GR. The theorem does not address:

- extended observers (e.g., networks of detectors),
- observers with nonlocal access to distributed information,
- hypothetical super-observers with knowledge beyond $J^-(\gamma)$.

Within classical GR, no physically plausible extension of observational capability circumvents the causal-past constraint, but the theorem is restricted to the local observer formulation.

F. Limitation to Coordinate-Time Reparametrization

The theorem identifies the reparametrization of the foliation label $t \rightarrow t' = f(t)$ and its associated lapse transformation as the source of temporal gauge freedom. It does not analyze more general refoliation symmetries involving changes in spatial coordinates or combined spacetime diffeomorphisms beyond the class that preserves the foliation structure. These general diffeomorphisms are well understood in GR, but isolating the temporal component requires the ADM setting used here.

G. No Claim About Relational Reformulations

Finally, the theorem does not propose an alternative formulation of physics without t . It merely establishes that the ADM foliation parameter cannot be observed. A relational or renewal-based formulation of dynamics, where evolution is indexed by invariant internal structure rather than external time, requires further development. Such formulations are the subject of subsequent papers.

VIII. CONCLUSION

This paper has established that in general relativity, the global temporal parameter used to label the slices of a foliation carries no operational meaning for an internal observer. By working entirely within the standard Lorentzian and ADM frameworks, we demonstrated that the lapse function $N(x, t)$ and the associated coordinate time t are gauge artifacts: they determine how spacetime is sliced and labeled, but they have no measurable effect on the experience or recorded history of any physical observer.

The central result, the Relativistic Temporal Non-Constructibility Theorem, shows that for any two foliations related by a smooth, strictly monotonic reparametrization, all empirical data accessible to an observer remain unchanged. Proper time along worldlines, defined geometrically by the metric and invariant under all diffeomorphisms, is the only operationally meaningful temporal quantity. The observable history of an observer depends solely on this invariant temporal structure and on the geometric events encountered along the worldline, not on the global slicing parameter used to evolve initial data.

This conclusion clarifies several long-standing features of general relativity. The Hamiltonian constraint

$$\mathcal{H} \approx 0 \quad (30)$$

and the vanishing of the canonical Hamiltonian are direct expressions of temporal gauge freedom. The timeless form of the Wheeler-DeWitt equation likewise reflects the fact that global time is not a physical degree of freedom. In cosmology, the standard choice of $N = 1$ defining “cosmic time” is a convenient gauge fixing rather than an observable structure inherent to the universe. Across all these contexts, this work identifies the underlying reason: slicing-based time coordinates do not correspond to operationally recoverable quantities.

The theorem provides a transparent bridge between the classical formulation of the Temporal Non-Constructibility Theorem, [2] proven in the context of autonomous dynamical systems, and the relativistic generalization presented here. In both settings, temporal parameters used to label the evolution of a system fail to correspond to measurable physical magnitudes. What is observable is the ordering and structure of events along worldlines, not the external parameter used to describe their progression.

By isolating the limits of time observability in classical general relativity, this work lays the foundation for a fully operational and relational account of temporal structure. The next steps extend these insights to renewal-based formulations of dynamics, to the emergence of proper time from distinguishability and entropy, and to the quantum and cosmological regimes where temporal gauge freedom becomes even more pronounced.

Global coordinate time serves as a useful descriptive tool, but it is not part of the physical content of the theory. Proper time, local, geometric, and invariant, constitutes

the only empirically accessible temporal structure. This is the heart of temporal non-constructibility in general relativity.

Appendix A: ADM Reparametrization Calculations

This appendix collects the explicit formulas underlying the lapse-reparametrization correspondence used in Section IV. All derivations are standard but included here for completeness and clarity. The results demonstrate how a smooth reparametrization of the foliation parameter induces a compensating transformation of the lapse function without altering the four-geometry.

1. ADM Decomposition and Proper Time

Let (M, g_{ab}) be a globally hyperbolic spacetime equipped with a foliation $M \cong \mathbb{R} \times \Sigma$ by spacelike hypersurfaces Σ_t . In adapted coordinates (t, x^i) , the metric takes the standard 3+1 form:

$$ds^2 = -N^2 dt^2 + h_{ij}(dx^i + N^i dt)(dx^j + N^j dt), \quad (\text{A1})$$

where:

- $N(t, x)$ is the lapse,
- $N^i(t, x)$ is the shift,
- $h_{ij}(t, x)$ is the induced spatial metric on Σ_t .

For an observer with four-velocity $u^a = (dx^a/d\lambda)$, the proper time increment is:

$$d\tau^2 = -g_{ab} u^a u^b d\lambda^2. \quad (\text{A2})$$

In particular, along a worldline with fixed spatial coordinates (for conceptual clarity), the metric yields:

$$d\tau = N dt. \quad (\text{A3})$$

2. Smooth Reparametrization of the Foliation Parameter

Consider a smooth, strictly monotonic reparametrization of the foliation label:

$$t \mapsto t' = f(t), \quad f'(t) > 0. \quad (\text{A4})$$

The coordinate differential transforms as:

$$dt = \frac{dt'}{f'(t)}. \quad (\text{A5})$$

The spatial coordinates x^i are unchanged, and the induced spatial metric h_{ij} is unaffected.

Under this transformation, the line element becomes:

$$ds^2 = -N^2 \left(\frac{dt'}{f'(t)} \right)^2 + h_{ij} \left(dx^i + N^i \frac{dt'}{f'(t)} \right) \left(dx^j + N^j \frac{dt'}{f'(t)} \right). \quad (\text{A6})$$

This can be rewritten in ADM form with a transformed lapse N' and shift N'^i .

3. Transformation of the Lapse

Comparing the transformed metric with the ADM expression:

$$ds^2 = -N'^2 dt'^2 + h_{ij}(dx^i + N'^i dt')(dx^j + N'^j dt'), \quad (\text{A7})$$

we identify the new lapse as:

$$N' = N \frac{dt}{dt'} = \frac{N}{f'(t)}. \quad (\text{A8})$$

Thus, *reparametrization of the coordinate time induces a rescaling of the lapse*:

$$\boxed{N' = N \frac{dt}{dt'}}. \quad (\text{A9})$$

The shift vector transforms similarly:

$$N'^i = \frac{N^i}{f'(t)}, \quad (\text{A10})$$

but in many contexts (e.g., comoving or normal slicing) $N^i = 0$, and the transformation reduces to the lapse scaling alone.

4. Invariance of the Four-Geometry

Crucially:

- The metric components expressed in (t', x^i) encode the same four-geometry as in (t, x^i) .
- All diffeomorphism-invariant objects (curvature tensors, causal structure, proper time, light cones) remain unchanged.
- The transformation $N \mapsto N'$ compensates exactly for the reparametrization of t , ensuring geometric equivalence.

Thus, the mapping:

$$(t, N) \mapsto (t', N') = \left(f(t), N \frac{dt}{dt'} \right) \quad (\text{A11})$$

is a gauge transformation of the ADM variables that leaves the physical content invariant.

5. Proper Time Independence

For a timelike worldline γ , proper time is given by:

$$d\tau^2 = -g_{ab} dx^a dx^b. \quad (\text{A12})$$

Because the four-metric g_{ab} is unchanged under the foliation reparametrization, it follows that:

$$d\tau' = d\tau. \quad (\text{A13})$$

Thus, proper time along worldlines is invariant under any transformation of the form $t \mapsto t' = f(t)$.

This invariance is the key step allowing the temporal non-constructibility theorem to generalize from autonomous classical systems (where orbit paths are invariant under reparametrization) to GR (where worldline histories are invariant under foliation reparametrization).

6. Summary

The ADM formalism makes the following facts explicit:

- i. Changing the foliation parameter t induces a rescaling of the lapse N .
- ii. This transformation leaves the four-geometry, causal structure, and proper time unchanged.
- iii. Therefore, *coordinate time and the lapse function are gauge variables*.
- iv. This reparametrization invariance plays the same structural role as reparametrization in autonomous classical flows, enabling the relativistic Temporal Non-Constructibility Theorem.

Appendix B: Explicit Examples in Standard Spacetimes

This appendix provides concrete demonstrations of the relativistic Temporal Non-Constructibility Theorem in familiar geometries. In each case, two different foliations or lapse choices are used to describe the same four-geometry. We show explicitly that an internal observer following a timelike worldline records the same physical observables and the same proper-time history, despite differences in coordinate time and lapse.

The examples are chosen to illustrate three canonical cases:

- i. Static black hole spacetimes (Schwarzschild)
- ii. Cosmological spacetimes (FLRW)
- iii. Flat spacetime (Minkowski and Rindler)

1. Schwarzschild Spacetime

(Static vs. Ingoing Eddington-Finkelstein Slicings)

The Schwarzschild metric in Schwarzschild coordinates is:

$$ds^2 = -\left(1 - \frac{2M}{r}\right) dt^2 + \left(1 - \frac{2M}{r}\right)^{-1} dr^2 + r^2 d\Omega^2. \quad (\text{B1})$$

Here the lapse is:

$$N = \sqrt{1 - \frac{2M}{r}}. \quad (\text{B2})$$

Now consider the ingoing Eddington-Finkelstein (IEF) foliation, defined by the change of coordinate:

$$v = t + r_*, \quad r_* = r + 2M \ln \left| \frac{r}{2M} - 1 \right|. \quad (\text{B3})$$

The metric becomes:

$$ds^2 = -\left(1 - \frac{2M}{r}\right) dv^2 + 2 dv dr + r^2 d\Omega^2. \quad (\text{B4})$$

In this slicing, the lapse is different and the foliation parameter has been changed, but:

- The spacetime events (the manifold points) are the same.
- The geometry is the same.
- A static observer at fixed r follows the same geometric worldline.
- Proper time increments satisfy

$$d\tau^2 = -g_{ab} dx^a dx^b, \quad (\text{B5})$$

and are independent of whether the observer uses t or v .

a. Conclusion for Schwarzschild. Even though the coordinate time and lapse differ radically between (t, r) and (v, r) :

- The observer's causal past $J^-(\gamma)$ is the same.
- The values of fields $\phi(\gamma(\tau))$ are the same.
- The recorded history lies in the same equivalence class.

This explicitly demonstrates the theorem in a strongly curved spacetime.

2. FLRW Cosmology

(Cosmic Time vs. Conformal Time)

A spatially flat FLRW universe has the metric:

$$ds^2 = -dt^2 + a(t)^2(dx^2 + dy^2 + dz^2), \quad (\text{B6})$$

with lapse $N = 1$.

Define conformal time via:

$$d\eta = \frac{dt}{a(t)}. \quad (\text{B7})$$

The metric becomes:

$$ds^2 = a(\eta)^2 (-d\eta^2 + dx^2 + dy^2 + dz^2), \quad (\text{B8})$$

with lapse:

$$N(\eta) = a(\eta). \quad (\text{B9})$$

Although cosmic time t and conformal time η differ by a nontrivial reparametrization, and the lapse changes accordingly, the following facts remain true:

- A comoving observer $(x, y, z) = \text{const}$ follows the same geometric worldline.
- Proper time satisfies

$$d\tau = dt, \quad (\text{B10})$$

independent of η .

- The observed redshifts, temperature of the CMB, and any measurable scalar fields give identical histories.

a. Conclusion for FLRW. Cosmic time and conformal time represent *different foliations of the same space-time*, and no internal observation can detect which slicing is the “real” one. This matches the theorem precisely.

3. Minkowski Spacetime

(Inertial vs. Rindler slicings)

In standard inertial coordinates:

$$ds^2 = -dt^2 + dx^2 + dy^2 + dz^2. \quad (\text{B11})$$

Now introduce Rindler coordinates:

$$t = \xi \sinh(\eta), \quad x = \xi \cosh(\eta), \quad (\text{B12})$$

which describe a uniformly accelerated observer.

The metric becomes:

$$ds^2 = -\xi^2 d\eta^2 + d\xi^2 + dy^2 + dz^2. \quad (\text{B13})$$

The lapse in Rindler time is:

$$N(\xi) = \xi. \quad (\text{B14})$$

Yet:

- Any inertial observer’s proper time satisfies $d\tau^2 = -ds^2$ and is invariant.
- Any accelerated observer’s proper time is computed from the same metric g_{ab} , regardless of whether we describe the geometry in (t, x) or (η, ξ) .
- The causal structure $J^-(\gamma)$ is identical.
- Observed histories of fields along γ are the same.

a. Conclusion for Minkowski. Even in flat spacetime, reparametrizing the time coordinate or choosing a non-inertial slicing does not change observable histories. This provides the simplest example of the theorem.

4. Summary of the Examples

Across Schwarzschild, FLRW, and Minkowski spacetimes:

- Different choices of foliation (different coordinate time functions).
- Different lapse functions.
- Different ADM decompositions.

all describe the *same geometry* and yield *the same internal observations* for any timelike observer.

These explicit examples demonstrate the central claim of the Relativistic Temporal Non-Constructibility Theorem:

Global coordinate time and the lapse function in GR are gauge variables; only proper-time evolution along worldlines has operational significance.

Appendix C: Technical Lemmas

This appendix collects the geometric facts required in the proof of the Relativistic Temporal Non-Constructibility Theorem. All statements follow from standard results in Lorentzian geometry and the ADM formulation of general relativity.

1. Invariance of Causal Pasts Under Foliation Reparametrization

Lemma 1 (Causal-Past Invariance). *Let (M, g_{ab}) be a smooth Lorentzian manifold, and let $\{\Sigma_t\}$ and $\{\Sigma_{t'}\}$ be two smooth foliations of M related by a strictly monotonic reparametrization*

$$t' = f(t), \quad f' > 0. \quad (\text{C1})$$

Then for any timelike worldline γ , the causal past of γ is invariant:

$$J^-(\gamma)_{\{t\}} = J^-(\gamma)_{\{t'\}}. \quad (\text{C2})$$

Proof. The causal past $J^-(\gamma)$ is defined as the set of all points $p \in M$ from which a future-directed causal curve reaches γ . This definition depends solely on the conformal and time-orientation structure of g_{ab} . A change of foliation parameter $t \mapsto t' = f(t)$ alters only the labeling of hypersurfaces; it does not alter the metric. Since causal relations are metric-invariant, the causal past is unchanged. $\square$

2. Proper-Time Invariance Under Lapse Rescaling

Lemma 2 (Proper-Time Invariance). *Let $\gamma(\lambda)$ be a time-like worldline in a spacetime (M, g_{ab}) . Let $\{\Sigma_t\}$ and $\{\Sigma_{t'}\}$ be foliations related by*

$$t' = f(t), \quad f' > 0, \quad (\text{C3})$$

with the lapse transforming as

$$N' = N \frac{dt}{dt'}. \quad (\text{C4})$$

Then the proper-time functional

$$\tau[\gamma] = \int \sqrt{-g_{ab} \frac{dx^a}{d\lambda} \frac{dx^b}{d\lambda}} d\lambda \quad (\text{C5})$$

has the same value in both foliations.

Proof. The proper-time functional is defined purely by the metric g_{ab} and the geometric path traced by γ . Since neither the path nor the metric changes under a reparametrization of the slicing, the integrand and integral are identical. $\square$

3. Invariance of Observable Fields Evaluated Along a Worldline

Lemma 3 (Field-Value Invariance). *Let $\phi : M \rightarrow \mathbb{R}$ be any smooth scalar field. Let $\gamma(\tau)$ be a worldline parametrized by proper time. If $t \mapsto t'$ is any smooth, strictly monotonic reparametrization of the foliation parameter, then the observable field history*

$$\Phi(\tau) = \phi(\gamma(\tau)) \quad (\text{C6})$$

is unchanged.

Proof. The scalar field ϕ assigns a value to each spacetime point, and the worldline $\gamma(\tau)$ passes through the same sequence of points regardless of foliation. Since proper time τ depends only on the geometry of the curve and not on the foliation parameter, the function $\Phi(\tau)$ remains the same. $\square$

4. Invariance of Observable Histories as Equivalence Classes

Lemma 4 (Path-Class Invariance). *Let*

$$\mathcal{H}_\gamma = \{\phi_1(\gamma(\tau)), \dots, \phi_k(\gamma(\tau))\} \quad (\text{C7})$$

be the vector of recorded scalar observables along a worldline γ . Representing this as a parametrized curve

$$\Gamma : \tau \mapsto (\phi_1(\gamma(\tau)), \dots, \phi_k(\gamma(\tau))) \in \mathbb{R}^k, \quad (\text{C8})$$

define the observable history as the equivalence class

$$[\Gamma] \in \text{Path}(\mathbb{R}^k) / \text{Diff}^+(\mathbb{R}). \quad (\text{C9})$$

Then for any smooth reparametrization of the foliation $t \mapsto t'$, the observable history class remains invariant.

Proof. From Lemmas 1–3:

- the same spacetime events lie on the worldline,
- proper time is unchanged,
- each scalar field takes the same values at those events.

Therefore the curve Γ in $\mathbb{R}^k$ is unchanged as a map of proper time. Thus its equivalence class under reparametrization is unchanged. $\square$

5. Summary

The lemmas above establish that under any smooth reparametrization of the foliation parameter:

- causal pasts are invariant,
- proper time along a worldline is invariant,
- scalar observables evaluated along that worldline are invariant,
- and the resulting observable history lies in the same reparametrization-equivalence class.

These results constitute the formal algebraic and geometric foundation for the Relativistic Temporal Non-Constructibility Theorem.

[1] R. Arnowitt, S. Deser, and C. W. Misner, *Gravitation: An Introduction to Current Research*, edited by L. Witten (Wiley, 1962) reprinted in *Gen. Relativ. Gravit.* 40, 1997 (2008).
[2] A. Riaz, The temporal non-constructibility theorem in classical dynamical systems (v1.0), Zenodo (2025).
[3] B. S. DeWitt, *Physical Review* **160**, 1113 (1967).
[4] R. M. Wald, *General Relativity* (University of Chicago

Press, 1984).
[5] S. W. Hawking and G. F. R. Ellis, *The Large Scale Structure of Space-Time* (Cambridge University Press, 1973).
[6] E.ourgoulhon, *3+1 Formalism in General Relativity*, Lecture Notes in Physics, Vol. 846 (Springer, 2012).
[7] P. A. M. Dirac, *Proceedings of the Royal Society A* **246**, 333 (1958).
[8] A. Einstein, *Annalen der Physik* **17**, 891 (1905).